LESSON-18

Functions...

i. Vertical Line Test: (More than 1 intersection: Not a function)

ii. Horizontal Line Test:

Many One (More than 1)

One One. (Intersecting only once)...

** FUNCTIONAL EQUATIONS :

1. $f(xy) = f(x) + f(y)$

   $\boxed{f(x) = k\log_a x}$

2. $f(xy) = f(x) \cdot f(y)$ then $\boxed{f(x) = x^n}$

3. $f(xy) = x f(y) + y f(x)$ then $f(x) = x\log_a x$.

4. $f(xy) = \frac{f(x)}{y} + \frac{f(y)}{x}$ then $f(x) = \frac{\log_a x}{x}$

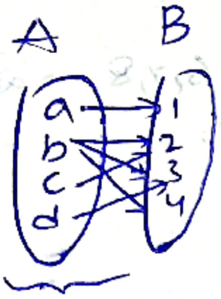
5. $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$ then $\boxed{f(x) = x^n}$

6. $f(f(x)) = kx - f(x) \Rightarrow f(x) = Ax + B$.

7. $f(x+y) = f(x) + f(y) \Rightarrow f(x) = f(1) \cdot x$

8. $f(x+y) = f(x) \cdot f(y) \Rightarrow f(x) = (f(1))^x$

9. $f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right) \Rightarrow \boxed{f(x) = 1 \pm x^n}$ ...



- Every Element in A R to Atleast one Element in B.

$f = \{(a, b)\ a \in A\ b \in B\}$

Domain CoDomain

Range $\leq$ CoDomain...

$f: R \to R$.

i. Many One

ii. Bijection. : Both injective, Surjective.

iii. injection (one-one) : $f(x_1) = f(x_2)$, $x_1 = x_2$ } Strictly Increase, Dec

iv. Surjection (Onto) : every element in codomain R has Preimage in Domain

v. Onto : Range = CoDomain

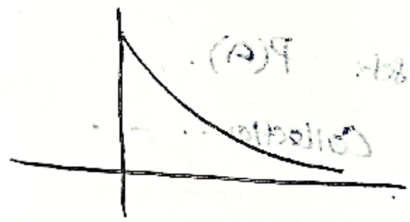
vi. One One. :

vii. Into : Range $\neq$ Co-Domain.

** Note : Range
Say $x \in [a,b]$

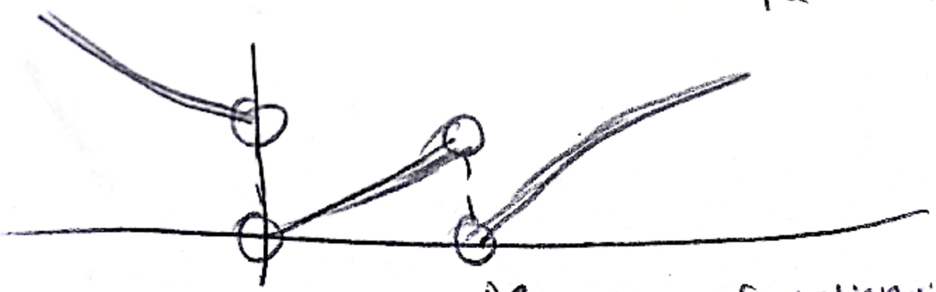
⇒ For a Strictly increasing function, decreasing Func

Minimum Point : a Max : a.
Max Point : b Minimum : b.



"Many one" :

Same y value for Different values of x & cut's Horizontal Multiple Times.



Many One function..

//

Domain calculation:

1. $\log_y(x) \Rightarrow x>0,\ y\neq 1$
2. $\sqrt{f(x)} \Rightarrow f(x)\geq 0$
3. $\frac{1}{g(x)} \Rightarrow g(x)\neq 0$

1. $f(x)=\sqrt{2-x}-\frac{1}{\sqrt{9-x^2}}$

$2-x\geq 0 \quad 9-x^2>0$

$2\geq x \quad 9>x^2$

$x\leq 2 \quad (3-x)(3+x)>0$

$x\in(-\infty,2],\ x\in(-3,3)$

$x\in(-3,2]$

$-3, 2 \quad 3$

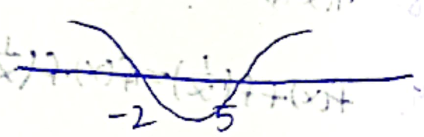
Domain Of Step:

1. Domain of: $\frac{[x]}{\sqrt{[x]^2-3[x]-10}}$

$[x]^2-3[x]-10>0$

$([x]-5)([x]+2)>0$

$(t-5)(t+2)>0$



$t\in(-\infty,-2)\cup(5,\infty)$

~~$t\in(-\infty,-3]\cup[6,\infty)$~~ $\quad t\in\{6,7,8,\dots\infty\}\cup\{-3,$

$x\in(-\infty,-2)\cup[6,\infty)$

4. $\sin^{-1}(h(x)) \Rightarrow -1\leq h(x)\leq 1$

Types of functions:

1. One-One: Every Element in Domain is Related in Exactly One Element is com

Ex: f (function) $R \to R$

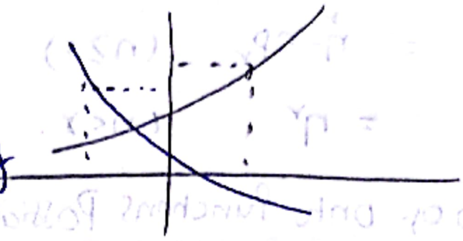
$f(x) = x^3$

$f'(x) = 3x^2$

(Streatly increasing function)..

or

(Streatly decreasing)

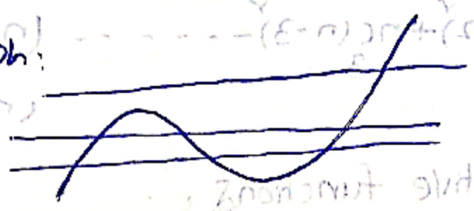


** $f'(x) > 0$ One-one.

$f'(x) \geq 0$, ≤ 0 } Not one one. Many one

Method - II

Graph:



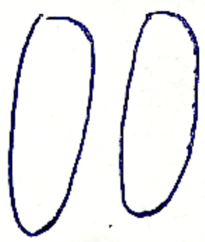
- Cutting only one time one one
- Cutting More than one time Many one or Same Y Value for Different Values of X

1. $f: R \to R$ $f(x) = \sin x$ $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$

$f'(x) = \cos x$

$= +ve.$

Hence one-one.

2.  Function : n^r

One-one :

2. Onto function:

let Set A contains 'r' elements $f: A \to B$ set B contains 'n' elements.

** 1. Total no. of functions Possiable $= n^r$

2. Total no. of one one functions $= {}^nP_r$ $(n \geq r)$

$= 0$ $(n < r)$

3. Total no. of Many One Functions Possiable

$= n^r - {}^nP_r \quad (n \geq r)$

$= n^r \quad (n < r).$

4. Total no. of onto functions Possiable :

$= n^r - {}^nC_1(n-1)^r + {}^nC_2(n-2)^r - {}^nC_3(n-3)^r + {}^nC_4(n-4)^r - - - - - - - - - - \quad (n \leq r)$

$= 0. \quad (n > r)$

5. Total no. of into functions.

$= {}^nC_1(n-1)^r - {}^nC_2(n-2)^r + {}^nC_3(n-3)^r - - - - - - \quad (n \leq r)$

$= n^r \quad (n > r)$

6. Total no. of Bijective functions.

$= n! \quad (n = r)$

$= 0. \quad (n \neq r)$

Odd even

$f(-x) = -f(x)$

$f(-x) = f(x)$

Periodic function.

$P(x+T) = P((x))$

$\sin px = \frac{2\pi}{p}$

$|\sin px| = \frac{\pi}{p} \quad |\sin x| = \pi$

9. Composite function :

$f(x) = \frac{2x+3}{3x+2}$ Also $x \neq \frac{-2}{3}$

$f(f(x)) = \frac{2f(x)+3}{3f(x)+2} = \frac{2\left(\frac{2x+3}{3x+2}\right)+3}{3\left(\frac{2x+3}{3x+2}\right)+2} = \frac{13x+12}{12x+13}$

$D \ldots x \neq \frac{-13}{12}$

Pilee

1. $a+\alpha=1 \quad b+\beta=2 \quad af(x)+\alpha f\left(\frac{1}{x}\right)=bx+\frac{\beta}{x} \quad x\neq 0$

then $\dfrac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$

$af(x)+\alpha f\left(\frac{1}{x}\right)=bx+\frac{\beta}{x}$

$af\left(\frac{1}{x}\right)+\alpha f(x)=\frac{b}{x}+\beta x$

Replace x with $\left(\frac{1}{x}\right)$

$$a\left(f(x)+f\left(\tfrac{1}{x}\right)\right)+\alpha\left(f\left(\tfrac{1}{x}\right)+f(x)\right)=b\left(x+\tfrac{1}{x}\right)+\beta\left(x+\tfrac{1}{x}\right)$$

$$f(x)+f\left(\tfrac{1}{x}\right)(a+\alpha)=x+\tfrac{1}{x}(b+\beta)$$

$$\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}=\frac{b+\beta}{a+\alpha}=\frac{2}{1}=2.$$

2.

$f(x)=\left(\dfrac{\sqrt{3e}}{2\sin x}\right)^{\sin^2 x}$: $\dfrac{K}{e}$ Max Value

then $\left(\frac{K}{e}\right)^8+\frac{K^8}{e^5}+K^8$

$\boxed{f'(x)=0.}$

$$\sin^2 x \log\left(\frac{\sqrt{3e}}{2\sin x}\right)$$

$$\sin^2 x\cdot\frac{2\sin x}{\sqrt{3e}}+\sin 2x\log\left(\frac{\sqrt{3e}}{2\sin x}\right)=0.$$

$$\frac{\cancel{2}\sin^3 x}{\sqrt{3e}}=-\cancel{2\sin x}\cos x\left(\log\frac{\sqrt{3e}}{2\sin x}\right)$$

$$\frac{\sin^2 x}{\sqrt{3e}}=-\cos x\left(\log\left(\frac{\sqrt{3e}}{2\sin x}\right)\right)$$

$$\Rightarrow y=\left(\frac{\sqrt{3e}}{2\sin x}\right)^{\sin^2 x}$$

$$\log y=\sin^2 x\left(\log\frac{\sqrt{3e}}{2\sin x}\right)$$

$$\log y=\sin^2 x\left[\log\left(\frac{\sqrt{3e}}{2}\right)-\log \sin x\right]$$

$$\frac{dy}{dx}\cdot\frac{1}{y}=2\sin x\cos x\left[\quad\right]+\sin^2 x\left[0-\frac{1}{\sin x}\cdot\cos x\right]$$

$(1,\infty)$, Jee Main 2024. $(-\infty,\infty)$

$R \to R$

$[0,\infty)$

3. $f(x) = \begin{cases} \log_e x & x>0 \\ e^{-x} & x\le 0 \end{cases}$, $g(x) = \begin{cases} x & x\ge 0 \\ e^{x} & x<0 \end{cases}$

$g(f(x)) = ?$

$g(f(x)) = \begin{cases} f(x) & f(x)\ge 0 \\ e^{f(x)} & f(x)<0 \end{cases}$

$g(f(x)) = \begin{cases} \log_e x & \log_e x \ge 0 \text{ and } x>0 \quad (x\ge 1) \\ e^{-x} & e^{-x}\ge 0 \text{ and } x\le 0 \\ e^{\log_e x} & \log_e x<0 \text{ and } x>0 \\ e^{e^{-x}} & e^{-x}<0 \text{ and } x\le 0 \end{cases}$

$e^{\log_e x^{-1}}$

$g(f(x)) = \begin{cases} \log_e x & x\ge 1 \longrightarrow [0,\infty) \\ e^{-x} & x\le 0 \longrightarrow (0,1] \\ \frac{1}{x} & 0<x<1 \longrightarrow (1,\infty) \end{cases}$

Same y value for Diff values of x

Hence: $\boxed{\text{Manyone}}$.

$gof: R \to R$. Range $= [0,\infty) \ne$ Codomain R

Hence: $\boxed{\text{Into}}$

Onto: For Range = Codomain.

$g(f(x)) = \begin{cases} f(x) & f(x)\ge 0 \\ e^{f(x)} & f(x)<0 \end{cases}$

$g(f(x)) = \begin{cases} e^{-x} & (-\infty,0) \\ e^{\log_e x} & (0,1) \\ \log_e x & [1,\infty) \end{cases}$